

Design and Implementation of a Single-Cycle RISC-V Processor

Project Report for CSC311 — Intro. to Computer Architecture

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Abstract—This report presents the design, implementation, and simulation of a complete single-cycle RISC-V processor. The processor supports R-type, I-type, load, store, and branch instructions. The design includes the instruction fetch, decode, ALU, register file, and memory units. Simulation results demonstrate correct execution of arithmetic, logical, memory, and branch instructions.

Index Terms—RISC-V, Single-Cycle CPU, Verilog, ALU, Register File, Instruction Fetch, Branch

I. INTRODUCTION

RISC-V is an open-source instruction set architecture (ISA) widely used in academic and industrial research. In this project, we implement a single-cycle CPU that executes a subset of the RV64I instructions. A single-cycle CPU completes each instruction in one clock cycle, simplifying control logic but requiring all operations to fit in a single cycle. The processor is implemented in Verilog and simulated using Vivado/XSim.

II. PROCESSOR DESIGN

A. Overview

The processor consists of the following main modules:

- **Program Counter (PC):** Holds the address of the current instruction.
- **Instruction Memory (IMEM):** Stores instructions and provides them to the fetch stage.
- **Control Unit:** Generates control signals based on the opcode and funct fields.
- **Register File:** Stores 32 general-purpose 64-bit registers.
- **ALU:** Performs arithmetic and logic operations.
- **Data Memory (DMEM):** Stores and loads data for load/store instructions.
- **Immediate Generator:** Generates sign-extended immediate values for I-type, S-type, and B-type instructions.

B. Instruction Fetch

The program counter increments by 4 for sequential instructions. For branch instructions, the PC selects either PC+4 or the branch target based on the Branch signal and ALU zero flag.

C. Instruction Decode

Instruction fields rd, rs1, rs2, opcode, funct3, and funct7 are extracted. The control unit generates RegWrite, ALUSrc, MemRead, MemWrite, MemToReg, Branch, and ALUCtrl signals. The register file reads values of rs1 and rs2 and provides them to the ALU.

D. ALU and Immediate Handling

The ALU performs addition, subtraction, logical AND, OR, XOR, shift-left, and shift-right operations. For instructions with immediate operands, ALUSrc selects the immediate instead of rs2. The immediate generator supports I-type (12-bit), S-type (12-bit), and B-type (13-bit) immediates.

E. Memory Access

For load instructions, DMEM reads data from the computed address and writes it back to the register file. For store instructions, DMEM writes data from the register file to the computed address. Addressing is byte-aligned and 64-bit wide.

F. Writeback

The writeback stage selects between the ALU result or memory read data based on MemToReg.

III. IMPLEMENTATION

The processor is implemented in Verilog with the following top-level modules:

- `cpu_full.v`: Full single-cycle CPU with load/store and branch support.
- `cpu_branch.v`: CPU with branch execution.
- `cpu_min.v`: Minimal ALU-only CPU for testing purposes.
- `imem.v`, `dmem.v`, `regfile.v`, `alu.v`, `control.v`, `immgen.v`, `pc.v`: Supporting modules.

Simulation testbench (`tb_cpu_isa.v`) verifies correct execution of arithmetic, logical, memory, and branch instructions.

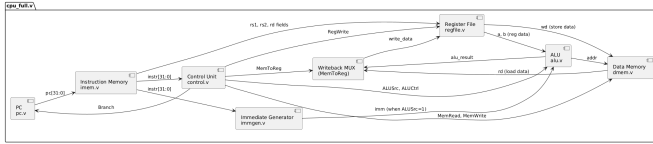


Fig. 1. Top-Level Block Diagram showing the integration of all CPU modules.

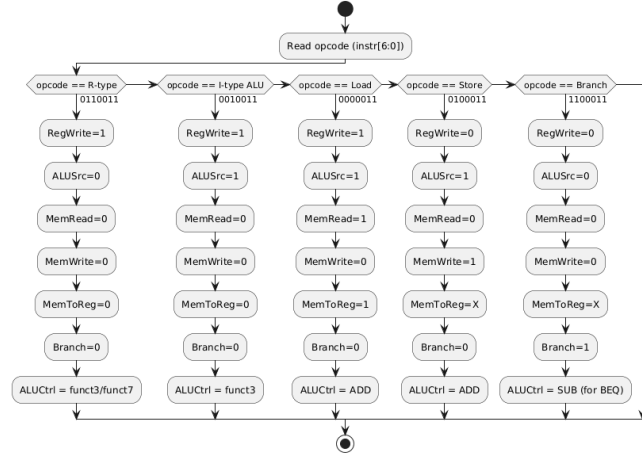


Fig. 2. Control Signal Decision Diagram: opcode mapping to RegWrite, ALUSrc, MemRead, MemWrite, MemToReg, Branch, and ALUctrl.

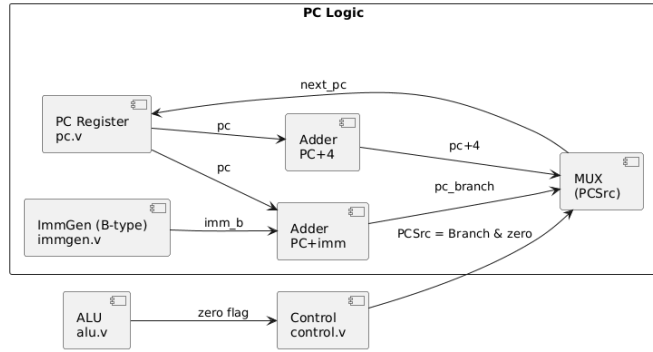


Fig. 3. Branch Datapath showing PC update logic based on Branch and Zero signals.

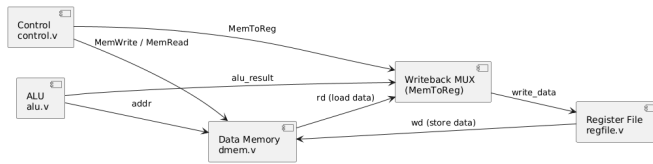


Fig. 4. Memory Access Diagram illustrating load and store operations in DMEM.

IV. DESIGN DIAGRAMS

V. SIMULATION AND RESULTS

Simulation was performed using Vivado XSim with a clock period of 10ns. The testbench initializes registers and runs a sequence of instructions, checking the final register values against expected results. Figures 5–9 show the waveform simulations for representative instructions.

A. Register File Results

All instructions executed correctly, producing the following values:

- x1 = 5, x2 = 12, x3 = 17, x4 = 7
- x5 = 4, x6 = 13, x7 = 9, x8 = 20
- x9 = 6, x10 = 40, x12 = 1, x13 = 4
- x14 = 13, x15 = 10, x16 = 17, x17 = 222

B. Instruction Waveforms

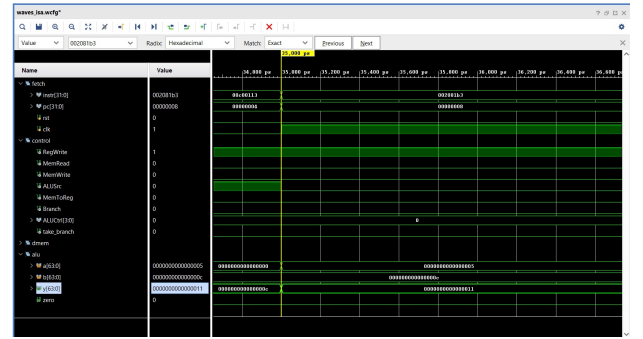


Fig. 5. Simulation waveform of R-type ADD: instr=002081b3, RegWrite=1, ALUSrc=0, MemToReg=0. ALU inputs: a=0x5, b=0xC → y=0x11.

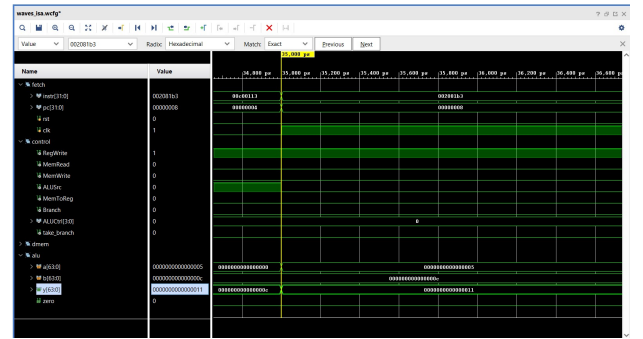


Fig. 6. Simulation waveform of LD (Load): instr=00003803, MemRead=1, MemToReg=1, RegWrite=1, reading rd=0x11 from address=0x0.

VI. CONCLUSION

We successfully designed and simulated a single-cycle RISC-V processor in Verilog. The processor correctly executes R-type, I-type, load, store, and branch instructions. Simulation results verified the functionality of the ALU, register file, memory, and branch logic. This project provides a solid foundation for understanding single-cycle CPU design and RISC-V instruction execution.

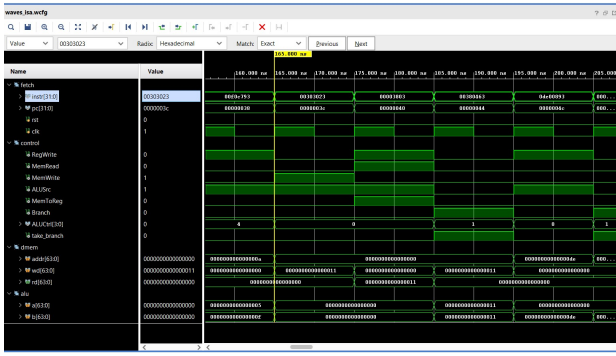


Fig. 7. Simulation waveform of SD (Store): instr=00303023, MemWrite=1 writes wd=17 to address 0x0.

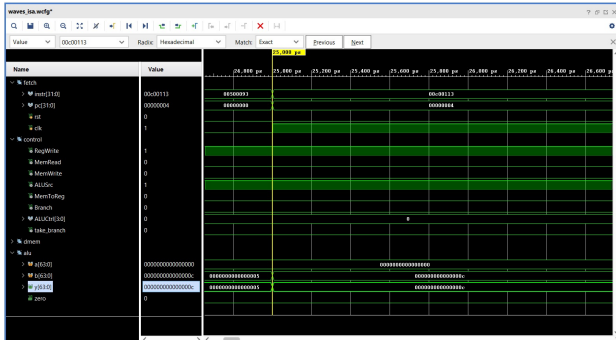


Fig. 8. Simulation waveform of I-Type ADDI: instr=00c00113, ALUSrc=1 selects immediate, RegWrite=1 updates x2=12.

ACKNOWLEDGMENT

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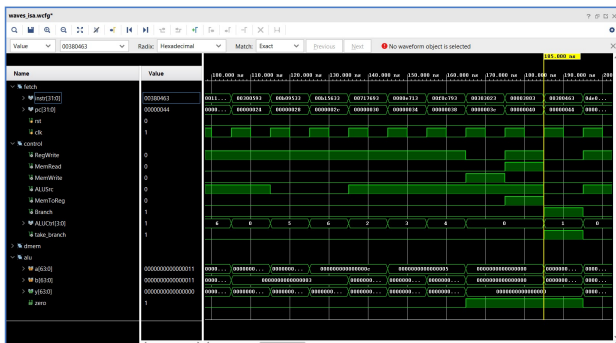


Fig. 9. Simulation waveform of BEQ (Branch): instr=00380463, Branch=1, take_branch=1, PC jumps to target.